



The Pricing of Derivatives using Volatility methods while integrating stochastic processes

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EDHEC X UNICA MASTER THESIS

SUPERVISOR : MIRCO RUBIN

DATE : 30/05/2025

Abstract

This research explores the application of stochastic processes in option pricing, aiming to enhance the traditional Black-Scholes framework. While the original model assumes constant volatility, this study introduces stochastic volatility to better capture the behavior of asset returns and improve pricing accuracy. Drawing on a range of academic and empirical sources, the analysis focuses particularly on the Heston model, evaluating its practical relevance, limitations, and potential for further development both in theoretical research and real-world trading environments.

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ACKNOWLEDGEMENTS

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Introduction

Option pricing is a fundamental key component of financial mathematics, providing an analytical framework for the valuation of derivatives. Since the appearance of the first models, the mathematical tools used to estimate the value of options have evolved progressively, in order to improve both their accuracy and their applicability. The historical foundation for this theory is the Black-Scholes model, a major breakthrough that introduced a rigorous method based on stochastic calculus and partial differential equations for valuing European options on non-dividend paying shares.

This model was not limited to its ability to determine a theoretical option price under certain assumptions: it also strongly impacted modern finance by establishing the foundations for subsequent developments. Over time, the limitations of Black-Scholes, notably the assumption of constant volatility and the exclusion of dividends, led researchers and practitioners to explore more realistic alternatives.

This search for improvement gave rise to several extensions of the original model: the inclusion of dividends in the Black-Scholes-Merton approach, the Cox-Ross-Rubinstein binomial tree for greater temporal flexibility, and the Heston model, which introduces a stochastic volatility framework to better capture observed market dynamics.

Today, option pricing theory remains a dynamic and evolving field, particularly with the increasing integration of stochastic models and more realistic assumptions about market behavior. This thesis explores this evolution through several key stages.

We begin with a literature review presenting the theoretical foundations of stochastic processes in finance and the transition from the Black-Scholes framework to more advanced models such as Heston's stochastic volatility model.

The second section develops the theoretical and methodological framework of the Heston model, detailing its mathematical underpinnings and implementation steps.

We then conduct an empirical analysis, where we implement both the Black-Scholes and Heston models in Python and compare their pricing outputs against market data using rigorous statistical techniques.

The thesis concludes with a discussion of results, highlighting the strengths and limitations of each model, and outlining future perspectives, including the potential of rough volatility models and practical considerations in real-world trading environments.

II- Literature Review

II-1 Theoretical Background

II-1-1 The Advantages of Stochastic Processes in Financial Modeling

Within the framework of financial mathematics, the need to model randomness accurately is fundamental. Financial assets such as stocks, commodities, or interest rates evolve over time in an unpredictable fashion. Prices fluctuate due to countless micro and macroeconomic factors, market sentiment, and exogenous shocks. In order to capture this inherent uncertainty, finance scholars and industry experts rely on stochastic processes, mathematical frameworks that describe the evolution of random variables over time.

Definition and Importance:

A stochastic process is a set of random variables $\{X_t\}_{t \geq 0}$ defined on a probability space and indexed by time, which represents the evolution of a system under uncertainty. In finance, X_t might represent the price of a risky asset at time t , the level of volatility, or even the interest rate.

The strength of stochastic processes is based on their ability to describe both the temporal dynamics and the random behavior of financial variables. Unlike deterministic models that provide a single predicted trajectory, stochastic models account for all possible future paths of a variable, each with its associated probability.

The application of stochastic processes in finance provides key advantages:

- It enables the construction of probabilistic models for asset price behavior.
- It forms the mathematical foundation for pricing and hedging derivative instruments.
- It allows for the quantification of risk and the development of dynamic trading and risk management strategies.

These processes are central to continuous-time finance, particularly in the modeling of derivative prices and risk-neutral valuation.

Types of Stochastic Processes in Finance:

Before introducing the foundational Black-Scholes model, it is crucial to understand the basic stochastic processes upon which modern option pricing theory is built.

1. Brownian Motion (Wiener Process):

The **standard Wiener process** W_t is the basis of stochastic calculus and financial modeling. It is a continuous-time stochastic process that satisfies the following properties:

- $W_0 = 0$
- W_t has independent and stationary increments
- For $0 \leq s < t$, the increment $W_t - W_s \sim N(0, t - s)$
- The paths of W_t are almost surely continuous but nowhere differentiable

In finance, it is commonly assumed that asset prices evolve according to a **Geometric Brownian Motion (GBM)**, which is derived from the Wiener process. The GBM is described by the following stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where:

- S_t is the asset price at time t
- μ is the drift term (expected return)
- σ is the volatility (standard deviation of returns)
- W_t is a standard Brownian motion

This model assumes log-normal distribution of asset prices, ensures non-negativity, and allows for a closed-form solution for European option prices, as developed in the Black-Scholes framework.

2. Ornstein–Uhlenbeck Process

The **Ornstein-Uhlenbeck (OU)** process is a mean-reverting process often used to model variables such as interest rates or stochastic volatility. Its dynamics are given by:

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

where:

- μ is the long-term mean
- $\theta > 0$ controls the speed of mean reversion

- σ is the volatility

This process reflects the economic insight that certain variables do not drift indefinitely, but rather tend to return to a long-run average value.

3. Poisson Process and Jump-Diffusion Models

To capture discontinuous movements in asset prices for example due to earnings announcements, crashes, or macroeconomic news, jump processes are introduced. The most classical is the Poisson process, which models the case of random jumps over time. In finance, the Merton jump-diffusion model combines a standard Brownian motion with a jump component:

$$dS_t = \mu S_t dt + \sigma S_t dW_t + J_t S_t dq_t$$

where:

- q_t is a Poisson process
- J_t represents the size of the jumps.

These models provide a more realistic description of market behavior, accounting for fat tails and sudden shocks.

4. Stochastic Volatility Models

Empirical studies show that volatility is not constant over time. Models such as the Heston model introduce stochastic dynamics for volatility itself, typically using an additional process V_t :

$$dS_t = \mu S_t dt + V_t S_t dW_t^S \quad \text{and} \quad dV_t = \kappa(\theta - V_t)dt + \xi V_t dW_t^V$$

- V_t : instantaneous variance of S_t
- κ : speed of mean reversion (how fast variance returns to its long-term mean)
- θ : long-term average level of variance
- ξ : volatility of the variance process (also called **vol-of-vol**)
- dW_t^V : Brownian motion driving the variance process (typically correlated with dW_t^S)

These models capture key stylized facts of financial markets, such as volatility clustering and the leverage effect.

The most classical and influential application of stochastic processes in finance is the Black-Scholes model, which assumes that the underlying asset follows a geometric Brownian motion. This model laid the foundation for analytical option pricing and inspired the development of more sophisticated models addressing its limitations.

II-1-2 From Black-Scholes to Modern Option Pricing Models

The Black-Scholes model marked the beginning of modern option pricing by providing a closed-form solution based on the assumption that the asset price follows a geometric Brownian motion with constant volatility. While it remains a basis for understanding derivatives, its simplifying assumptions, particularly constant volatility and frictionless markets, limit its accuracy in capturing real market dynamics.

Over the past decades, this has motivated the development of more advanced models that aim to address these limitations and better fit observed option prices. The most prominent extensions include:

1. Stochastic Volatility Models:

Models such as Heston introduce a stochastic process for volatility itself. Instead of assuming a constant σ , these models allow volatility to fluctuate over time, often through a mean-reverting square-root process. This helps capture:

- Volatility smiles and skews
- The leverage effect (negative correlation between price and volatility)
- Clustering of volatility

The SABR model is another example, popular in fixed income and FX markets, where both the asset and its volatility follow correlated stochastic processes.

2. Rough Volatility Models:

Recent studies suggest that volatility is even more irregular than previously assumed, exhibiting roughness rather than smooth stochastic behavior. Rough volatility models, such as the Rough Heston model, extend the classical Heston framework by introducing fractional Brownian motion with a Hurst parameter $H < 0.5$ into the variance process. This modification captures:

- Short-term implied volatility smiles
- High-frequency volatility clustering
- More accurate calibration to market data, particularly for short maturities

3. Jump-Diffusion and Multifactor Models:

To model abrupt price changes, jump-diffusion models introduce discontinuous movements via Poisson processes. This extension better reflects empirical return distributions with fat tails and asymmetries.

Multifactor models go further by incorporating multiple sources of risk (e.g., stochastic interest rates, multiple volatility factors), enhancing flexibility but increasing computational complexity.

Empirical research shows that:

- The **Black-Scholes model** often underestimates the market prices of deep out-of-the-money options
- **Heston and SABR** models improve fit significantly, especially in the presence of volatility skew
- **Rough volatility models** often outperform classical stochastic volatility models on short maturities
- **Jump models** are particularly effective for assets subject to discontinuous price events.

Overall, modern pricing models offer a more realistic framework, but their complexity requires careful calibration and may lack closed-form solutions.

II-1-3 The Black-Scholes-Merton Model and Its Early Limitations:

The Black-Scholes-Merton (BSM) model provides the first closed-form solution for pricing European-style options. It represents a major achievement in financial economics, offering a mathematically structured approach built upon stochastic calculus and arbitrage theory. The model is based on the assumption that the underlying asset follows a geometric Brownian motion, and that markets operate under idealized, frictionless conditions.

Assumptions of the BSM Model:

The derivation of the Black-Scholes formula relies on the following key assumptions:

1. The asset price S_t follows a continuous diffusion process: $dS_t = \mu S_t dt + \sigma S_t dW_t$

where:

- μ is the **constant drift** (expected return)

- σ is the **constant volatility** of the asset
 - W_t is a **standard Brownian motion** (Wiener process)
2. Markets are frictionless: No transaction costs, taxes, or bid-ask spreads.
 3. Continuous trading: Investors can adjust their portfolios at any instant in time.
 4. Short-selling and borrowing/lending at a constant risk-free rate r are allowed without restriction.
 5. No dividends are paid on the underlying asset during the option's life.
 6. Volatility and interest rates are constant and known.
 7. The market is arbitrage-free and complete, allowing for perfect replication of option payoffs.
 8. European options only: exercise is allowed only at maturity.

Under these assumptions, the price of a European call option at time t , with strike price K and maturity T , is given by:

$$C(S_t, t) = S_t N(d_1) - K e^{r(T-t)} N(d_2)$$

with:

$$d_1 = [\ln(S_t/K) + (r + 1/2 \sigma^2)(T - t)] / \sigma \sqrt{T - t} \text{ and } d_2 = d_1 - \sigma \sqrt{T - t}$$

The formula for the European **put option** is obtained via put-call parity or directly derived:

$$P(S_t, t) = K e^{r(T-t)} N(-d_2) - S_t N(-d_1)$$

This model provides a benchmark for option pricing and has influenced decades of research and financial practice.

While the Black-Scholes-Merton model remains a cornerstone of modern financial theory, its practical application is significantly hindered by a set of simplifying assumptions that rarely hold in real-world markets. These assumptions, while essential to deriving the analytical pricing formula, introduce distortions that can lead to substantial pricing errors and ineffective risk management strategies.

One of the most fundamental assumptions of the BSM model is that volatility is constant over time. However, empirical studies consistently show that market volatility is both time-varying and state-dependent. It tends to cluster, exhibits mean-reversion, and may experience sudden jumps. Moreover, implied volatility, which is inferred from market prices of options, varies systematically with strike price and maturity, forming well-documented structures such as the volatility smile and skew. The Black-Scholes model, which assumes a flat volatility surface, fails to capture these phenomena, resulting in systematic mispricing, especially for deep in- or out-of-the-money options.

Another major limitation lies in the assumption of constant interest rates. In reality, interest rates fluctuate over time and can differ across maturities, introducing valuation biases in longer-dated derivatives. Similarly, the assumption that the underlying asset does not pay dividends restricts the model's applicability to a narrow set of assets, and while dividend adjustments can be applied externally, they lack internal consistency within the BSM framework.

The model also relies on the idealization of frictionless markets, where there are no transaction costs, no bid-ask spreads, and unlimited liquidity. In practice, such conditions do not exist. Traders face execution costs, and portfolio rebalancing is not continuous, particularly in times of market stress. These frictions generate hedging errors and undermine the effectiveness of dynamic replication strategies upon which the BSM model is built.

Further, the assumption of continuous trading and perfect liquidity suggests that market participants can adjust their positions instantaneously and without limitation. This is not only unrealistic in general but particularly problematic in the pricing of exotic or path-dependent derivatives. In times of low liquidity or crisis, markets deviate sharply from this assumption, exposing investors to risks not captured by the model.

In addition, the Black-Scholes model assumes that asset returns follow a log-normal distribution, effectively ruling out sudden, discrete price jumps. However, financial markets are frequently impacted by abrupt events such as earnings announcements or geopolitical news, which lead to discontinuities in asset prices. The inability of the BSM framework to incorporate such jumps limits its relevance in environments characterized by high uncertainty.

Finally, the model is designed exclusively for European-style options, which can only be exercised at maturity. This restricts its application to a subset of instruments and excludes many real-world contracts with early exercise features, such as American options, callable bonds, or barrier options.

In sum, while the Black-Scholes-Merton model remains a powerful theoretical tool, its strict assumptions limit its practical realism. These limitations justify the emergence of stochastic volatility models, such as Heston, which better capture the dynamics of volatility and improve pricing performance for real-world derivatives.

II-1-4 Stochastic Volatility Models: The Heston Framework

To overcome the limitations of the Black-Scholes model, most notably the assumption of constant volatility, researchers developed models in which volatility itself becomes a stochastic process. Among these, the **Heston model**, introduced by Steven Heston in 1993, stands out as one of the most widely used due to its balance between empirical accuracy and analytical tractability.

In the Heston model, the asset price S_t and its instantaneous variance V_t evolve according to a system of stochastic differential equations (under the risk-neutral measure)

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{V_t S_t} dW_t^S \\dV_t &= \kappa(\theta - V_t)dt + \xi V_t dW_t^V \\dW_t^S dW_t^V &= \rho dt\end{aligned}$$

Where:

- r is the constant risk-free interest rate,
- V_t is the variance process (i.e. σ_t^2)
- $\kappa > 0$ is the speed of mean reversion,
- $\theta > 0$ is the long-run mean of variance,
- $\xi > 0$ is the volatility of volatility (vol-of-vol),
- $\rho \in [-1, 1]$ is the correlation between the Brownian motions for the asset and its variance.

1. Stylized Facts and Intuition Behind the Model

One of the most important contributions of the Heston model is its ability to replicate several key empirical occurrences observed in financial markets. Unlike the Black-Scholes framework, which assumes constant volatility, the Heston model introduces stochastic volatility by allowing the instantaneous variance of the asset to evolve randomly over time. This captures the phenomenon of **volatility clustering**, where high-volatility periods tend to persist and alternate with calmer market phases. The model further incorporates the **leverage effect**, a negative correlation between asset returns and volatility innovations, which reflects the observed tendency for volatility to increase following sharp declines in asset prices.

Another major empirical strength of the Heston model lies in its ability to produce **volatility smiles and skews** in implied volatility surfaces.

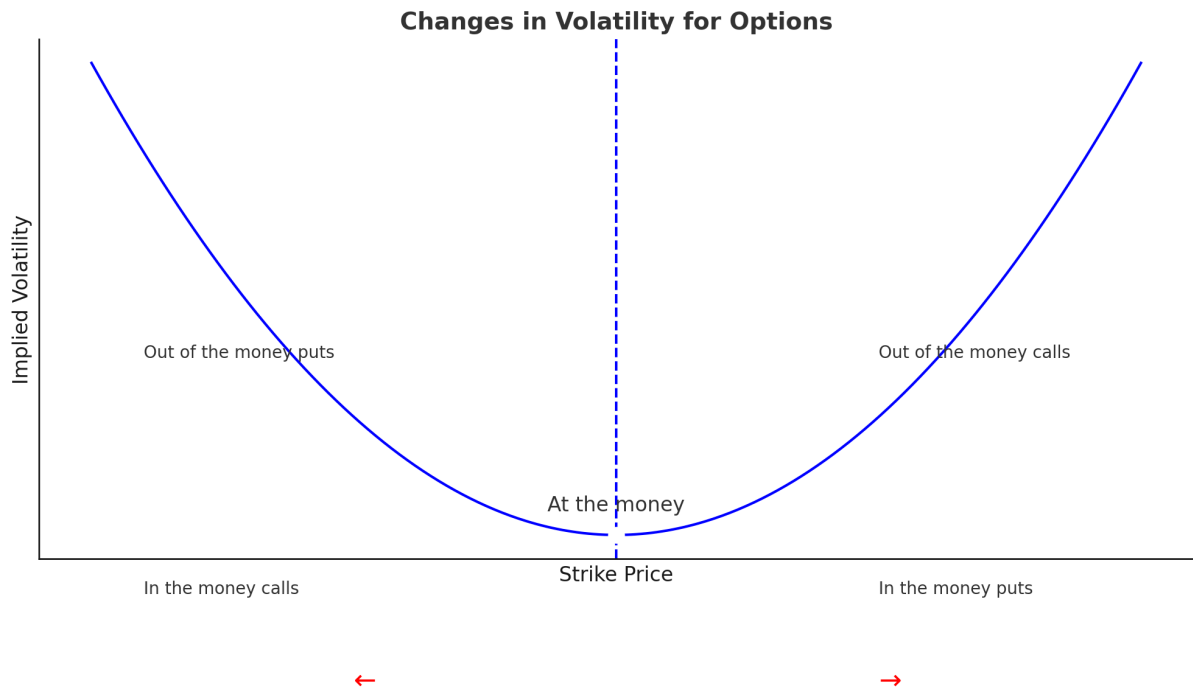


Figure showing the implied volatility smile for options

The graph shows implied volatility (vertical axis) as a function of the strike price (horizontal axis) for options with the same expiration date.

At the center are at-the-money (ATM) options, meaning the strike price is close to the current price of the underlying asset.

On either side, we observe:

- To the left: Out-of-the-money (OTM) puts (put options with a strike price well below the spot), exhibiting higher implied volatility.
- To the right: Out-of-the-money (OTM) calls (call options with a strike price well above the spot), also with higher implied volatility.

At the bottom of the curve, ATM options have the lowest implied volatility.

This phenomenon, known as the volatility smile, challenges the Black-Scholes assumption of constant volatility. It suggests that the market anticipates a higher likelihood of extreme outcomes, which is reflected in a volatility premium for deep in-the-money (ITM) or out-of-the-money (OTM) options.

This feature, absent in the Black-Scholes model, is critical for matching the pricing patterns seen in option markets across different strikes and maturities.

Finally, the model ensures **mean reversion of variance**, meaning that volatility does not drift arbitrarily but fluctuates around a long-term average, a realistic feature especially relevant for long-dated options.

2. Analytical Tractability and Pricing Formula

Despite its increased complexity, the Heston model remains analytically tractable. It allows for a semi-closed-form solution for the price of European options using Fourier transform techniques, specifically through the characteristic function of the log-price. This makes the model efficient to implement numerically, while still capturing the stochastic nature of volatility. The call price, for instance, can be expressed in terms of two probability-like quantities P_1 and P_2 , which are derived from integrals of the characteristic function. The mathematical derivation, usually presented in appendices, involves advanced techniques from complex analysis but results in a usable formula suitable for calibration and risk management.

3. Practical Implementation and Calibration Challenges

In practice, the Heston model has become a benchmark tool for pricing and hedging derivatives in various markets, including equities, foreign exchange, and volatility indices. However, its practical implementation comes with certain challenges. The model introduces five parameters κ , θ , ξ , ρ , and the initial variance V_0 which must be estimated from market data. This calibration process is typically done by minimizing the squared error between observed and model-implied option prices or volatilities. While more demanding than the Black-Scholes model, this procedure provides a significantly better fit to observed market prices.

4. Limitations and Further Developments

Despite its strengths, the Heston model is not without limitations. It assumes continuous price paths and does not incorporate jumps or discontinuities, making it less suited to assets that experience frequent shocks or macroeconomic surprises. In addition, although it captures the general shape of the implied volatility surface, it may struggle to fully match the slope of short-term skews or extreme kurtosis in asset returns. These residual discrepancies have motivated the development of more advanced frameworks, such as rough volatility models or jump-diffusion extensions.

Nevertheless, the Heston model remains one of the most widely used and studied models in financial engineering, striking a robust balance between realism and tractability. Its capacity to capture essential market dynamics while offering analytical solutions has made it an indispensable tool for both academic research and practical applications in quantitative finance.

II-2-Gap in the Literature:

Despite the significant advancements brought by stochastic volatility models such as Heston's, a gap remains between theoretical option pricing outputs and actual market observations. In particular, while these models succeed in capturing volatility smiles and leverage effects, they sometimes fall short in matching the finer details of implied volatility surfaces, especially at very short maturities or in turbulent market conditions.

This discrepancy highlights the limitations of smooth volatility models and has drawn attention to the need for frameworks capable of reproducing the irregular, fractal-like behavior of observed volatility. Empirical research has shown that volatility exhibits not only persistence and mean reversion, but also roughness, characterized by sample paths that are less regular than those generated by classical Brownian motion.

As a result, recent literature has focused on rough volatility models, which assume that volatility evolves according to a process with Hölder regularity lower than that of Brownian motion. These models offer improved calibration performance, especially for short-dated options, and provide a more accurate depiction of high-frequency market behavior.

This persistent mismatch between theoretical predictions and market data motivates ongoing research into more sophisticated pricing frameworks. The current work contributes to this literature by examining the relative performance of classical models like Black-Scholes, more advanced models like Heston, and highlighting the path toward integrating rough volatility dynamics into practical financial applications.

III-Theoretical and Methodological Framework

This derivation is based on the key article by Steven L. Heston, in which he proposes a closed-form solution for pricing options under stochastic volatility: **Heston, S. L. (1993).** *A closed-form solution for options with stochastic volatility with applications to bond and currency options*. *The Review of Financial Studies*, 6(2), 327–343 [\[8\]](#)

A more comprehensive and detailed derivation than the original one proposed by Heston (1993) is provided in the appendix.

This version reproduces the key steps of the author's reasoning while integrating additional insights drawn from personal research, as well as intermediate justifications and clarifications. The goal is to enhance accessibility and clarity, turning the derivation into a pedagogical resource designed to support a deeper understanding of the model.

III-1 Foundations of the Heston Model

Heston, S. L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options*. *The Review of Financial Studies*, 6(2), 327–343, p. 328. [\[8\]](#)

The Heston model extends the classical Black-Scholes framework by incorporating stochastic volatility, allowing the variance of the underlying asset to fluctuate randomly over time. The price dynamics of the underlying asset follow the stochastic differential equation:

$$dS(t) = \mu S dt + \sqrt{v(t)} S dz_1(t)$$

where:

- $z_1(t)$ is a standard Brownian motion,
- $v(t)$ represents the instantaneous variance of the underlying asset
- μ is the drift rate.

Unlike Black-Scholes, where the variance is constant, here it evolves according to its own dynamics. The square root of this variance is assumed to follow an Ornstein-Uhlenbeck-type process:

$$d\sqrt{v(t)} = -\beta \times \sqrt{v(t)} dt + \delta dz_2(t)$$

This equation captures the mean-reverting behavior of the volatility process, where β represents the speed of reversion and δ the intensity of fluctuations. The process remains stationary, ensuring that

volatility does not diverge over time. Furthermore, the introduction of two correlated Brownian motions, $z_1(t)$ and $z_2(t)$, through a correlation coefficient ρ , establishes a dependency between the dynamics of the asset price and its volatility.

III-2 Application of Itô's lemma and variance dynamics

Heston, S. L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options. The Review of Financial Studies*, 6(2), 327–343, p. 329. [\[8\]](#)

To determine the dynamics of y , apply Itô's lemma to the function $f(x) = y^2$

where: $y(t) = \sqrt{v(t)}$

This change of variable means that the variance can be expressed as the square of a simpler stochastic variable. The calculation gives :

$$dv(t) = [\delta^2 - 2\beta\sqrt{v(t)}] + 2\delta\sqrt{v(t)}dz_2(t)$$

We then rewrite this equation in the more classical form introduced by Cox, Ingersoll and Ross (CIR):

$$dv(t) = [k(\theta - v(t))]dt + \sigma\sqrt{v(t)}dz_2(t)$$

avec $\sigma = 2\delta$, $k = 2\beta$ et $\theta = \delta^2$. This representation ensures that $v(t)$ remains positive and defines a realistic model for the instantaneous variance.

III-3-Valuation equation and absence of arbitrage

Heston, S. L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options. The Review of Financial Studies*, 6(2), 327–343, p. 332. [\[8\]](#)

In a no-arbitrage framework, the value of a derivative $U(s,v,t)$ must satisfy a partial differential equation (PDE) that incorporates the dynamics of S and v , the correlation between S and v , and the price of risk. This equation is :

$$\begin{aligned} & (1/2) \cdot v \cdot S^2 \cdot (\partial^2 U / \partial S^2) + \rho \cdot \sigma \cdot v \cdot S \cdot (\partial^2 U / \partial S \partial v) + (1/2) \cdot \sigma^2 \cdot v \cdot (\partial^2 U / \partial v^2) + r \cdot S \cdot (\partial U / \partial S) \\ & + [K \cdot [\theta - v(t)] - \lambda(S, v, t)] \cdot (\partial U / \partial v) - r \cdot U + (\partial U / \partial t) = 0 \end{aligned}$$

The term $\lambda(S, v, t)$ represents the price of volatility risk. Breeden (1979) shows that this risk can be related to the aggregate consumption of a representative investor, giving a microeconomic basis to the valuation equation.

III-4-Boundary conditions and general form of the solution

Heston, S. L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options. The Review of Financial Studies*, 6(2), 327–343, p. 334. [\[8\]](#)

The solution to the stochastic volatility model for a European call option is derived from the equation introduced in the previous section.

$$C(S, v, t) = SP_1 - kP(t, T)P_2$$

This formulation is a direct extension of Black-Scholes, where P_1 and P_2 are adjusted risk-neutral probabilities, depending on v and t . The boundary conditions are defined by the behaviour of price S , volatility v and maturity T , guaranteeing the existence of a unique solution to the PDE.

III-5- Solving by Fourier transform and affine form

Heston, S. L. (1993). *A closed-form solution for options with stochastic volatility with applications to bond and currency options. The Review of Financial Studies*, 6(2), 327–343, p. 334. [\[8\]](#)

The method for solving the Heston model is based on the Fourier transform and the Feynman-Kac theorem. We show that the characteristic functions associated with the probabilities P_1 and P_2 can be written in affine form:

$$f_j(x, v, t, \phi) = e^{C_j(T-t; \phi) + D_j(T-t; \phi)v + i\phi x}$$

This affine structure allows efficient Fourier inversion to obtain C_j and D_j , and therefore the option price. The explicit expressions for C_j and D_j follow directly from the PDE, and depend on the model parameters $(k, \theta, \sigma, \rho, \lambda)$. (See appendix for the full expression of C_j and D_j).

Numerical integration allows these probabilities to be calculated, making the Heston model both theoretically sound and numerically implementable, offering a robust and more realistic alternative to the Black-Scholes model for options pricing.

IV-Experimentation / Empirical Analysis

IV-1-Methodology

IV-1-1-Intuition

To gain an initial intuition about the expected results, we decided to conduct our first tests on a small data sample. Based on the characteristics observed in this preliminary dataset, we then analyzed a larger set of data to verify our hypotheses.

We compared the call and put option prices obtained using the Black-Scholes and Heston models with actual market prices in order to determine whether stochastic volatility indeed leads to a significant improvement in our results.

Once the call option prices are established, we performed a test on the mean of the residuals relative to zero.

For the various current market option prices, we used the **Nasdaq website**, which provides the latest option prices along with their different strike levels for various tickers.

IV-1-2-Origin of the data

To conduct an initial comparative test between the Black-Scholes and Heston models, we begin by analyzing a small, controlled dataset. The objective is to observe whether Heston's stochastic volatility model provides more accurate pricing for short-term options, and to use this observation as the empirical motivation for the hypothesis we aim to test more rigorously later.

For this preliminary exercise, we collected option data from the NASDAQ website on May 14, 2025, covering call and put options on Tesla stock (TSLA) with a uniform expiration date of May 16, 2025 i.e., three calendar days after data collection. The file was exported in **.xlsx** format and imported into Python for further analysis.

The raw dataset includes 13 columns, divided between call and put options. For each type, the available variables are:

- Expiration Date
- Last Traded Price

- Change
- Bid Price
- Ask Price
- Volume
- Open Interest
- Strike Price

To ensure data quality and relevance, we applied the following filters:

- Only options expiring on May 16, 2025 were retained to guarantee a consistent maturity.
- All contracts with missing, zero, or negative values for key variables (Bid, Ask, Spot price, Implied Volatility) were excluded.
- We also removed illiquid options, i.e., those with zero Bid/Ask or zero volume (when volume data was available).

After applying these criteria, the final sample consisted of 25 option contracts, exclusively focused on call options, which were then used to compute relative pricing errors for both models. This filtered dataset serves as the foundation for the first step in our empirical analysis, which informs the formulation of our working hypothesis: that the Heston model outperforms the Black-Scholes model in pricing accuracy under stochastic volatility conditions.

Below is an example of a table obtained for call options on Tesla stock:

Calls							
Exp. Date	Last	Change	Bid	Ask	Volume	Open Int.	Strike
May 16	32.00	15.40	31.90	32.80	1286	3186	287.50
May 16	29.87	14.87	29.60	30.50	4445	8583	290.00

Table showing call option data for Tesla stock as of May 16, 2025.

We then introduce python code to enable us to implement our Black-Scholes and Heston models.

IV-2-Implementation of the Black Scholes model in Python

IV-2-1-Importing volatility

Before starting to implement the Black-Scholes model, we need to calculate the implied volatility of our underlying asset.

To do so :

1. Download closing prices : Over the last 500 trading days from yfinance.
2. Calculation of daily logarithmic returns :

$$r_t = \ln\left(\frac{p_t}{p_{t-1}}\right)$$

3. Calculation of the standard deviation of these daily yields :

$$\sigma_{daily} = std(r_1, \dots, r_n)$$

It is an empirical average of the dispersion of returns, so :

$$\sigma_{daily} = \frac{1}{n-1} \sum_{i=1}^n (r_i - \bar{r})^2$$

4. Annualisation: 252 trading days/year are assumed, so :

$$\sigma_{annualized} = (\sigma_{daily}) \times 252$$

IV-2-2-Black-Scholes model

We then compute our Black-Scholes model in Python using the previously calculated volatility (the code is provided in the appendix).

We obtain the results in the following format:

Strike	Bid	Ask	BS_Price
--------	-----	-----	----------

287.5	31.90	32.80	31.22814 5
290.0	29.60	30.50	28.83717 9

Table showing the call option prices calculated using the Black-Scholes method for different strike prices on Tesla stock on 13/05/2025 with a maturity of 3 days

IV-3-Heston Model in Python

We design a Python program implementing the Heston model (see appendix), and we thus derive prices for our puts and calls based on different strike prices.

We then obtain the following table:

<i>Strike</i>	<i>Bid</i>	<i>Ask</i>	<i>Heston_Call</i>
287.5	31.90	32.80	31.754156
290.0	29.60	30.50	29.482113

Table showing the call option prices calculated using the Heston method for different strike prices on Tesla stock on 13/05/2025 with a maturity of 3 days

IV-4-Understanding Our Analysis Method: B-S vs Heston

IV-4-1-General Idea

In order to evaluate the performance of the Black-Scholes and Heston models in estimating option prices, we adopt a quantitative approach based on the analysis of relative errors. For each call option, we compare the theoretical value calculated by the models with the market midpoint, defined as the average of the Bid (buy) price and the Ask (sell) price. This midpoint is considered a reliable approximation of the market "fair price" because it lies between buying and selling intentions.

The formula used to calculate the relative error is as follows:

$$\text{Relative error} = ((\text{Model Price} - \text{Market Midpoint}) / \text{Market Midpoint}) \times 100$$

This approach makes it possible to neutralize the effect of price scale and to compare the models' performance more consistently, regardless of the option price level.

We then perform a univariate Student's t-test on the series of relative errors of each model. The objective of this test is to determine whether the average error is statistically different from zero. The null hypothesis H_0 states that the mean of the relative errors is zero, meaning the model is unbiased. On the contrary, the alternative hypothesis H_1 states that the mean of the errors is significantly different from zero, which would indicate a systematic bias of the model (overestimation or underestimation of prices).

The test is performed with a particularly strict significance level:

$$\alpha = 0.01\% = 0.0001$$

This means that we accept a risk of only 0.01% of wrongly concluding that the model is biased when it is not (type I error).

We obtain the following results with Excel:

Mean error Heston	Mean error B-S	STD error Heston	STD error B-S	nb values	T-value Heston	T-value B-S	p-value Heston $\alpha=0,01\%$	p-value B-S $\alpha=0,01\%$
5,553945199	28,63224714	6,361288767	22,13170341	25	4,365424525	6,468604473	0,000208565	1,08906E-06

Table comparing the P-values for the Heston and Black-Scholes models on the price deviations of call options calculated using these different methods for various strikes on Tesla stock on 13/05/2025 with a maturity of 3 days

Thus, in the case of options on Tesla shares dated 13/05/2025 with a strike expiring on 16/05/2025, we conclude that the null hypothesis H_0 is accepted for the Heston model, but rejected for the Black-Scholes model. This suggests that the Heston model provides a better fit for this dataset, yielding more accurate call option price estimates than Black-Scholes.

IV-4-2-Limitations

One of the main challenges encountered in this study lies in the rigorous comparison of the performance of the Black-Scholes and Heston models in estimating option prices. Indeed, a great number of factors influence the price of an option, making the overall evaluation of a model particularly complex. The mere fact that the underlying asset, maturity, and strike price can vary already generates a massive number of combinations. If we consider an infinite number of possible values for these three dimensions, theoretically, this leads to a combinatorial explosion of possible configurations across the three dimensions.

To this must be added a fundamental element: unlike Black-Scholes, the Heston model relies on several structural parameters that need to be fixed or calibrated, namely:

- κ : the mean reversion speed
- θ : the long-term variance
- σ : the volatility of volatility
- ρ : the correlation between the asset price and its variance

This calibration requirement introduces a new level of complexity: for each combination of underlying / maturity / strike, one could consider recalibrating these parameters to obtain the best possible approximation of the market price. However, such an approach, although rigorous, would lead to an endless analysis, hardly compatible with the practical constraints of an academic research thesis.

IV-5-Methodological Choice

IV-5-1-Data choice

To gain a broader and more representative view of model performance, we extended our analysis to a significantly larger dataset. This second dataset includes 1951 option contracts (calls and puts combined), with maturities ranging from a few days to several months, and covering underlying assets from various industries to enhance sectoral diversity.

The data was sourced from **OptionMetrics** and imported in **.xlsx** format under the file name *option call*. The original dataset included the following variables:

- date: date of observation
- act_symbol: ticker symbol of the underlying asset
- expiration: option maturity date
- strike: strike price
- call_put: option type (Call or Put)
- bid, ask: bid and ask quotes
- vol: implied volatility
- delta, gamma, theta, vega, rho: standard option Greeks

For consistency and focus, we decided to retain only call options. This choice is motivated by the put-call parity principle, which enables the reconstruction of put prices based on call prices, the spot level, and the risk-free rate.

The put-call parity formula used is:

$$P=C-S+K \cdot \exp(-rT)$$

Where:

- P is the price of the put
- C is the price of the call
- S is the spot price of the underlying asset
- K is the strike price
- r is the risk-free interest rate
- T is the time to maturity (in years)

We then have **951** data points. Thanks to OptionMetrics, we directly obtained implied volatilities (vol) for each selected underlying.

This allowed us to use these implied volatilities without requiring further calibration. The corresponding column in the dataset was therefore retained and cleaned for direct use in model computations. To complete the dataset, we retrieved the spot prices of all underlyings as of February 9, 2019, using Yahoo Finance.

These spot prices were matched with the options based on ticker and date to allow for consistent pricing comparisons under both the Black-Scholes and Heston models.

This comprehensive dataset provides a robust framework for testing model performance across a wide range of strikes, maturities, and underlying assets, making it suitable for a rigorous and statistically meaningful empirical evaluation.

IV-5-2-Choice of the parameters : Calibration / Validation

In the following section, we choose to use the first **2/3** of our dataset to determine the optimal values for the Heston model parameters: κ , θ , σ , and ρ .

To this end, we develop a Python script (provided in the appendix) that systematically tests various combinations of these parameters and identifies the one that minimizes the mean relative error between the model-predicted and market-observed option prices.

Using this calibration procedure, we obtain the following optimal parameter values:

$$\mathbf{kappa=10.00, theta=0.00, sigma=1.00, rho=-0.50}$$

This combination yields a mean relative error of **63.90%**.

This approach allows us to identify parameter values that best fit our dataset as a whole. It also enables a fair comparison between the Black-Scholes and Heston models using the remaining one-third of the data, which is reserved for out-of-sample evaluation.

We also set $r=0.02625$, which corresponds to the risk-free rate observed on 1-year U.S. Treasury bill [\[9\]](#)

IV-5-3-Limits with these parameters

Although the calibration procedure identifies a parameter set that minimizes the mean relative error, some of the values, particularly $\theta = 0.00$, raise theoretical concerns. A long-term variance of zero implies that the variance process is expected to decay completely in the long run, which contradicts observed market behavior and theoretical models of persistent volatility.

Nevertheless, this result can be partially justified by the **short maturity** of the options in our dataset. Since most contracts expire within a few days, the influence of long-term variance dynamics on option pricing is relatively minor. In this context, the short-term behavior of the variance process dominates the pricing outcome, and the model effectively reduces to a highly mean-reverting stochastic volatility model with an initially high variance but no sustained volatility level.

V-Results

Mean error BS	Mean error Heston	STD error BS	STD error Heston	nb valu es	T-val ue BS	T-value Heston	p-value BS a=0,001%	p-value Heston a=0,001%
---------------------	-------------------------	--------------------	------------------------	------------------	----------------	-------------------	------------------------	-------------------------------

56,196 82	44,12985	102,27 95	52,74368	325	9,905 236	15,083 55	0%	0%
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Table comparing the P-values for the Heston and Black-Scholes models on the call price deviations calculated using these different methods for various strikes on equities dated 09/02/2019.

This table shows that the mean relative error of the Heston model is lower than that of the Black-Scholes model (44.13% vs. 56.20%), and the standard deviation of the errors is also smaller under Heston (52.7% vs. 102.3%). This suggests that the Heston model provides more accurate and consistent pricing than the Black-Scholes model across our dataset.

However, both models yield p-values close to 0, indicating that the null hypothesis of a mean relative error equal to zero is rejected in both cases. In other words, neither model perfectly replicates market prices.

This discrepancy is not surprising. Our objective was not to find a model with zero pricing error, but rather to compare the overall performance of both pricing frameworks. To do so, we deliberately constructed a dataset that includes a broad spectrum of options: various maturities, out-of-the-money contracts, and some with very low premium values close to zero. Under such conditions, pricing becomes particularly challenging, and both models are expected to deviate more significantly from observed market prices.

Overall, the Heston model appears to outperform Black-Scholes in this diverse testing environment, demonstrating better adaptability to complex option characteristics.

VI-Discussion

The results obtained confirm that the Heston model outperforms the Black-Scholes model in estimating option prices. Thanks to its ability to incorporate stochastic volatility, the Heston model provides valuations that are closer to the prices observed on the market, and therefore potentially more reliable for investors.

This improved accuracy has practical implications. In theory, it should encourage practitioners to favor Heston for option pricing, particularly when identifying arbitrage opportunities or constructing

more precise hedging strategies. Indeed, a more realistic modeling of the underlying asset price dynamics allows for a better assessment of potential discrepancies between theoretical value and market price.

However, despite its advantages, the Heston model also presents several limitations:

- Its computational complexity remains a barrier to widespread adoption. Unlike Black-Scholes, which offers a simple and fast closed-form solution, Heston requires the numerical evaluation of complex integrals (via characteristic functions), making it more time-consuming and resource-intensive.
- Its calibration is delicate: to obtain meaningful results, it is often necessary to adjust the model parameters for each underlying asset, or even for each option, which makes its implementation more technical.
- Finally, while the Heston model introduces stochastic volatility, it does not incorporate “rough volatility”. This more recent approach captures volatility variations at very short time scales (microstructure) and appears to better reflect the empirical behavior of volatility on financial markets. From this perspective, Heston remains a limited approximation of reality.

VII-Conclusion

VII-1-Conclusion

We have shown that introducing stochastic processes to model volatility significantly improves option pricing accuracy compared to the constant-volatility Black-Scholes model. In particular, stochastic volatility better reflects market behaviors, especially through the empirical shape of the implied

volatility surface. This leads to more accurate option valuation, especially for **out-of-the-money** options, where the Black-Scholes model often deviates considerably from observed market prices.

However, this improvement in precision comes at the cost of increased complexity. While the Heston model remains elegant and analytically tractable under certain conditions, it is more computationally demanding than Black-Scholes. Its implementation requires advanced numerical methods (such as Fourier transforms or Monte Carlo simulations) and a careful calibration process to be effective in practice.

Moreover, it is important to note that our findings are based on a sample of approximately 900 representative yet limited data points. Therefore, the conclusions drawn should not be considered universally applicable, but rather as trends observed within this specific sample. It remains essential to test these results against other market data and refine them in various contexts.

VII-2-Limitations and Perspectives

Despite the advances introduced by the Heston model, it still fails to capture certain fine-grained volatility behaviors, particularly micro-level short-term variations (microstructure noise) and the roughness of volatility observed in high-frequency data. Recent empirical studies have shown that volatility trajectories do not follow smooth processes as assumed in classical models, but instead exhibit rough behavior characterized by low regularity.

In this context, the Rough Heston model emerges as a promising extension. In this model, volatility follows a fractional process, often modeled using a fractional Brownian motion with a Hurst exponent $H < 0.5$. This framework captures more accurately:

- the short-term temporal dependence of volatility,
- the persistence of certain shocks,
- and a more realistic empirical structure, particularly in high-frequency markets.

VII-3-Further Perspectives

While the Heston model offers a rigorous and analytically tractable framework for stochastic volatility in option pricing, its practical adoption in trading floors remains limited. During several interviews with professionals and experts in the field, a consistent observation was highlighted: while stochastic volatility models are mathematically robust, their real-world use, particularly for vanilla option pricing in front-office environments, is still not widely adopted compared to local volatility models, such as the one proposed by Dupire.

Indeed, in most trading environments, especially in equity derivatives and volatility desks, the local volatility framework is favored due to its operational simplicity and direct calibration to the market. Unlike stochastic models, which require the estimation of hidden state variables and are sensitive to initial conditions and parameter instability, the Dupire model constructs a deterministic volatility surface that exactly fits observed implied volatilities across all strikes and maturities. This property is particularly appreciated by traders and structurers because it ensures pricing consistency across books, facilitates real-time hedging, and meets regulatory demands for transparency and robustness.

Moreover, the deterministic nature of the local volatility surface allows for faster and more stable computation of Greeks, which are central to hedging strategies. In practical terms, this translates into smoother PnL attribution, easier integration into risk management systems, and greater confidence in the model's output, especially under stressed market conditions. In contrast, the implementation of the Heston model often involves numerical approximations, complex calibration procedures, and sensitivity to parameter choice, which can limit its adoption in production environments.

Nevertheless, stochastic volatility models retain their value in certain contexts. They are often used in model validation, exotic option pricing, or for stress-testing purposes where capturing the empirical dynamics of volatility, such as clustering and the leverage effect, is essential. More recently, hybrid approaches have emerged, combining the strengths of both frameworks. Stochastic-Local Volatility (SLV) models, for example, seek to preserve the exact fit of the Dupire model while embedding realistic stochastic features in the volatility evolution. These models offer a promising compromise between theoretical realism and practical applicability.

Taking these perspectives into account, it appears that the Heston model, although commonly used in academic studies, remains perceived by many practitioners as too heavy for direct use in day-to-day trading. Its complexity, calibration requirements, and numerical cost often outweigh its benefits in environments where speed, transparency, and robustness are paramount.

Thus, future research could focus on the development of models that are not only faithful to observed market dynamics but also operationally viable in high-frequency environments. Improving calibration techniques, reducing computational time, and integrating stochastic volatility features into local volatility structures represent promising avenues to reconcile academic modeling innovations with the practical needs of trading floors.

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Appendices

Black and Scholes

In a risk-neutral world, a risk-free portfolio has a rate of return equal to the risk-free rate.

What must this portfolio contain?

- The option whose price we want to evaluate
- The underlying asset allowing to cancel the risks related to holding this option

Choose the risk-free portfolio:

- Selling a Call
- Buying delta shares

$$\pi = -C + \Delta S$$

Variation of the portfolio π :

$$d\pi = -dC + \Delta dS$$

With :

$$C = \text{Call}(S, t)$$

$$dS = \mu S dt + \sigma S dW$$

For the Call variation we take the first-order derivative with respect to dt while going up to the second order for dS .

$$dC = \frac{\partial C}{\partial t} dt + \frac{\partial C}{\partial S} dS + \frac{1}{2} \frac{\partial^2 C}{\partial S^2} dS^2$$

This is because the underlying asset's dynamics involve a stochastic term represented by dW , which requires a second-order derivative.

However, the time component, dt , is deterministic and only requires a first-order derivative.

In fact ,

$$dS^2 = \mu^2 \cdot S^2 \cdot dt^2 + 2 \cdot \mu \cdot S \cdot dt \cdot \sigma \cdot S \cdot dW + \sigma^2 \cdot S^2 \cdot dW^2$$

2nd Ordre

3/2 Ordre

1st Ordre

because $dW^2 = dt$

Let's recall that dC can be written in the form :

$$dC = \frac{\partial C}{\partial t} dt + \frac{\partial C}{\partial S} dS + \frac{1}{2} \frac{\partial^2 C}{\partial S^2} dS^2$$

And by replacing dS² we obtain:

$$dC = \frac{\partial C}{\partial t} dt + \frac{\partial C}{\partial S} dS + \frac{1}{2} \cdot \sigma^2 \cdot S^2 \cdot \frac{\partial^2 C}{\partial S^2} dt$$

We reintroduce this formula into the expression of dπ:

$$d\pi = - \left(\frac{\partial C}{\partial t} dt + \frac{\partial C}{\partial S} dS + \frac{1}{2} \cdot \sigma^2 \cdot S^2 \cdot \frac{\partial^2 C}{\partial S^2} dt \right) + \Delta dS$$

We simplify then the expression by introducing $\Delta = (\partial C / \partial S)$:

$$d\pi = - \left(\frac{\partial C}{\partial t} + \frac{1}{2} \cdot \sigma^2 \cdot S^2 \cdot \frac{\partial^2 C}{\partial S^2} \right) \cdot dt$$

By considering that the portfolio is risk-free and evolves at the risk-free interest rate, we can write:

$$d\pi = \pi \cdot r \cdot dt = (-C + \Delta S) \cdot r \cdot dt$$

$$d\pi = - \left(\frac{\partial C}{\partial t} + \frac{1}{2} \cdot \sigma^2 \cdot S^2 \cdot \frac{\partial^2 C}{\partial S^2} \right) \cdot dt$$

$$- \left(\frac{\partial C}{\partial t} + \frac{1}{2} \cdot \sigma^2 \cdot S^2 \cdot \frac{\partial^2 C}{\partial S^2} \right) \cdot dt = (-C + \Delta S) \cdot r \cdot dt$$

By rearranging this equation, we obtain the Black-Scholes equation:

$$\boxed{\frac{\partial C}{\partial t} + r \cdot S \cdot \frac{\partial C}{\partial S} + \frac{1}{2} \cdot \sigma^2 \cdot S^2 \cdot \frac{\partial^2 C}{\partial S^2} = r \cdot C}$$

Proof/Derivation of the Heston Stochastic Volatility Model

We assume as a starting hypothesis that the spot price of an asset follows the following diffusion process:

$$dS(t) = \mu S dt + \sqrt{v(t)} S dz_1(t)$$

With $Z_1(t)$ being a Brownian motion,

i.e.:

- $W_0=0$
- $W_s - W_t$ are independent for $s \neq t$
- $\forall s < t, W_t - W_s \sim \mathcal{N}(0; t-s)$
- $\mathbb{E}(W_t^2) = \text{Var}(W_t) = t$

We assume that the volatility follows an **Ornstein-Uhlenbeck (OU) process**, which is a stationary stochastic process that models a volatility with a tendency to revert to a mean.

$$d\sqrt{v(t)} = -\beta \times \sqrt{v(t)} dt + \delta dz_2(t)$$

In fact OU means that :

$$dx_t = \beta(\mu - x_t) dt + \sigma dW_t$$

with :

- dx_t : the infinitesimal drift of x_t at time t
- $\beta(\mu - x)$: the mean-reversion component that drives the process back toward its long-term equilibrium level
- β : the speed of mean reversion
- μ : the long-term mean, the value around which the process fluctuates
- dt : an infinitesimal time increment
- σ : the volatility (controls the magnitude of random fluctuations)
- dW_t : a Wiener process introducing randomness

Thus, in the Heston model:

- $\sqrt{v(t)}$: square root of the instantaneous variance at time t
- $d\sqrt{v(t)}$: infinitesimal evolution of the volatility
- β : speed of reversion to 0
- δ : the volatility of volatility
- $dz_2(t)$: a Wiener process

Then, by applying Itô's lemma, we show that the variance follows the following process:

$$dv(t) = [\delta^2 - 2\beta\sqrt{v(t)}] + 2\delta\sqrt{v(t)}dz_2(t) \quad (1)$$

As a reminder, Itô's lemma states that for X_t a process defined as:

$$dx_t = \varphi(t, x_t) + \sigma(t, x_t) dW_t$$

where:

- $\varphi(t, x_t)$: is the **drift** (trend term)
- $\sigma(t, x_t)$: is the **volatility** (diffusion term)
- W_t : is a **Wiener process**

If $f(t, X_t)$ is a function of class $\mathcal{C}^{1,2}$, meaning it is once differentiable with respect to t and twice differentiable with respect to X_t , the differential of f is given by:

$$df(t, x_t) = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma \frac{\partial f}{\partial x} dw_t$$

In our case, we apply Itô's lemma to the function $f(x) = x^2$, since $W(t) = (\sqrt{v(t)})^2$.

Letting $y(t) = \sqrt{v(t)}$, we have:

$$v(t) = f(y(t)) = y(t)^2$$

and:

$$dy(t) = -\beta y(t) dt + \delta dz_2(t)$$

We now apply Itô's lemma to $f(y(t)) = y(t)^2$, which yields:

$$df(y(t)) = 2y(t)dy(t) + \frac{1}{2}d^2y(t)^2$$

because $f''=2$

With:

$$2y(t)dy(t) = 2\sqrt{v(t)} \left(-\beta\sqrt{v(t)} dt + \delta dz_2(t) \right) = -2\beta v(t) dt + 2\delta\sqrt{v(t)} dz_2(t)$$

and,

$$(dy(t))^2 = \delta^2 dt \quad \text{because} \quad dz_2^2 = dt \quad (\text{properties of Wiener process})$$

By substitution, we obtain exactly equation (1).

→ Equation (1) can be reformulated in the well-known Cox-Ingersoll-Ross (CIR) structure, which is widely used in stochastic volatility modeling:

$$dv(t) = [k(\theta - v(t))] dt + \sigma\sqrt{v(t)} dz_2(t)$$

By identifying it with:

$$\delta^2 - 2\beta v(t) = k(\theta - v(t))$$

$$\text{And } 2\delta\sqrt{v(t)} = \sigma\sqrt{v(t)}$$

Done :

$$\sigma = 2\delta$$

$$k = 2\beta$$

$$\theta = \delta^2 / \beta = \sigma^2 / (4k)$$

With $z_1(t)$ correlated with $z_2(t)$ via a correlation coefficient ρ . For the sake of simplicity at this stage, we assume a constant interest rate r .

→ Discounting formula:

$$e^{-rh}$$

Starting again from the Black-Scholes-Merton no-arbitrage argument, we have that the value of any asset $U(s,v,t)$ must satisfy the following differential equation:

$$\begin{aligned} & (1/2) \cdot v \cdot S^2 \cdot (\partial^2 U / \partial S^2) \\ & + \rho \cdot \sigma \cdot v \cdot S \cdot (\partial^2 U / \partial S \partial v) \\ & + (1/2) \cdot \sigma^2 \cdot v \cdot (\partial^2 U / \partial v^2) \\ & + r \cdot S \cdot (\partial U / \partial S) \\ & + [K \cdot [\theta - v(t)] - \lambda(S, v, t)] \cdot (\partial U / \partial v) \\ & - r \cdot U \\ & + (\partial U / \partial t) \\ & = 0 \end{aligned} \quad (6)$$

Where $\lambda(S,v,t)$ represents the market price of volatility risk. In financial markets, volatility itself is uncertain and thus volatile. This uncertainty constitutes a risk, and every risk carries a price for investors.

Breeden (1979) stated that $\lambda(S,v,t) = \gamma \text{Cov}((dv/dc)/c)$, where $c(t)$ is the consumption rate and γ is the investor's coefficient of relative risk aversion.

→ Cox, Ingersoll, and Ross explained that:

$$\frac{dC(t)}{C(t)} = \mu v(t) C(t) dt + \left(\sigma \sqrt{v(t)} C(t) dz_3(t) \right)$$

Remark: In fact, Heston initially introduces a revised stochastic differential equation for SSS, incorporating a continuously evolving variance. The expression for $v(t)$ is then derived using Itô's lemma.

The objective is to generalize the Black-Scholes framework, so we return to the B-S equation, now applied to a function $U(s,v,t)$ that depends explicitly on the stochastic volatility v .

To ensure that $U(s,v,t)$ admits a unique solution, we need to specify how to value the volatility risk premium term $\lambda(s,v,t)(\partial U / \partial v)$.

To determine this risk price $\lambda(s,v,t)$, we introduce a more fundamental economic assumption: this is where Breeden's consumption-based model comes into play.

→ A European call option with strike K and maturity T satisfies equation (6), and is therefore subject to the following boundary conditions:

$$U(S, v, T) = \max(0, S - K) \quad (9)$$

$U(0, v, t) = 0$ (This is intuitive: if the underlying is worth 0, there is no chance the strike K will be exceeded)

$$\frac{\partial U}{\partial S}(\infty, v, t) = 1 \quad (\text{Makes sense since as } S \rightarrow \infty, \text{ the option is deep in the money})$$

$$rS \frac{\partial U}{\partial S}(S, 0, t) + k\theta \frac{\partial U}{\partial v}(s, 0, t) - rU(s, 0, t) + U_t(S, 0, t) = 0 \quad (\text{Thanks to (6) when } v(t) = 0)$$

$U(S, \infty, t) = S$ (When volatility is extremely high, the likelihood of the option expiring in the money approaches 1, so the option's value tends toward the price of the underlying asset)

→ By analogy with Black-Scholes, we deduce a solution of the form:

$$C(S, v, t) = SP_1 - ke^{-r(T-t)}P_2$$

Recall that the Black-Scholes solution is given by:

$$C(s, t) = SN(d_1) - ke^{-r(T-t)}N(d_2)$$

In essence, in our similar solution, P1 and P2 play roles analogous to N(d1) and N(d2), but they are significantly more complex in form. They are typically obtained by integrating over all possible paths of v(t), often using a Fourier transform.

→ Then we inject $C(S, v, t) = SP_1 - ke^{-r(T-t)}P_2$ in equation (6). But to simplify the calculations and create a simple expression, we impose: $x = \ln(S)$.

In fact in (6), we had at the origin:

$$S^2 \frac{\partial^2 U}{\partial S^2}; S \frac{\partial U}{\partial S} \quad \text{These terms are complex.}$$

$x = \ln(S)$ allow to simplify:

$$S \frac{\partial U}{\partial S} \rightarrow \frac{\partial U}{\partial x}$$

and

$$S^2 \frac{\partial^2 U}{\partial S^2} \rightarrow \frac{\partial^2 U}{\partial x^2} - \frac{\partial U}{\partial x}$$

→ After this change of variable and substitution in (6), we have:

$$\frac{1}{2} v \frac{\partial^2 P}{\partial x^2} + \rho \sigma V \frac{\partial^2 P_j}{\partial x \partial v} + \frac{1}{2} \sigma^2 V \frac{\partial^2 P_j}{\partial v^2} + (r + u_j V) \frac{\partial P}{\partial x} + (a_j - b_j \cdot v) \cdot \frac{\partial P_j}{\partial v} + \frac{\partial P_j}{\partial t} = 0$$

with $j = 1 \text{ or } 2$

and $u_1 = 1/2, u_2 = -1/2, a = k\theta, b_1 = k + \lambda - \rho\sigma, b_2 = k + \lambda$

Why are we doing this ?

In fact we just need to replace and factorise S and K.

Thus in (6) we obtain:

$$S \left[(1/2) v \frac{\partial^2 P_1}{\partial x^2} + \rho \sigma v \frac{\partial^2 P_1}{\partial x \partial v} + (1/2) \sigma^2 v \frac{\partial^2 P_1}{\partial v^2} + (n + 1/2) v \frac{\partial P_1}{\partial x} \right.$$

$$\left. + (k\theta - (k + \lambda - \rho\sigma) v) \frac{\partial P_1}{\partial v} + \frac{\partial P_1}{\partial t} \right] -$$

$$K \left[(1/2) v \frac{\partial^2 P_2}{\partial x^2} + \rho \sigma v \frac{\partial^2 P_2}{\partial x \partial v} + (1/2) \sigma^2 v \frac{\partial^2 P_2}{\partial v^2} + (n_2 - 1/2) v \frac{\partial P_2}{\partial x} \right.$$

$$\left. + (k\theta - (k + 1) v) \frac{\partial P_2}{\partial v} + \frac{\partial P_2}{\partial t} \right]$$

We have the same expression in both factorisation, we just need then to create a general P_j depending of u , b and a .

→ According to (9) we want $U(S, v, T) = \max(S - K, 0)$

So we want $C(S, v, T) = S P_1(x, v, T) - K P_2(x, v, T) = \max(S - K, 0)$

with $C(S, v, T)$: the price of the call

For $S > k$ and $(x > \ln(k))$,

$$C = S - k = S_1 - k \times 1, \text{ thus } P_1 = 1, P_2 = 1$$

For $S \leq k$ and $(x \leq \ln(k))$,

$$C = 0 = S \times 0 - k \times 0, \text{ thus } P_1 = 0, P_2 = 0$$

→ In this sense : $P_j(x, v, T, \ln(K)) = 1_{\{x > \ln(k)\}}$, for $j=1,2$

According to Cox and Ross, P_1 and P_2 can be interpreted as risk-neutral probabilities, adjusted to reflect the market's expectation under a no-arbitrage framework. Cox and Ross add that if we consider that x has the following stochastic process :

$$dx(t) = [r + u\sqrt{v}] dt + \sqrt{v}(t)dz_1(t)$$

$$dv = [a + b\sqrt{v}] dt + \sigma\sqrt{v}(t)dz_2(t)$$

with the parameters u , a et b defined above.

Then P is the conditional probability that the option expire la probabilité conditionnelle que l'option expire avec un strike équivalent :

$$P(x, v, T | \ln(K)) = P(x(T) \geq \ln(K) | x(t) = x, v(t) = v) \quad (15)$$

It comes from the Feynman-Kac theorem, which tells us that for X_t a diffusion such that:

$$dX = \mu(X, t) dt + \sigma(X, t) dW$$

and if $u(x, t)$ a function that satisfies the partial differential equation (6) in our case:

$$du/dt + \mu(x, t) du/dx + (1/2) \sigma(x, t)^2 \partial^2 u / \partial x^2 = 0$$

with $u(x, T) = f(x)$, then the solution to this equation is given by:

$$u(x, t) = E[f(X) | X = x] \text{ which is equal to (15) in our case}$$

→ The probabilities P_1 and P_2 are not immediately observable. Nevertheless we can find their characteristic function:

$f(x, v, T, \phi)$ check the conditions (12) and so,

$$f_j(x, v, T, \phi) = e^{i\phi x}$$

where $\phi \in \mathbb{R}$ is a Fourier variable.

As a reminder, the characteristic function is defined as:

$$\chi(\phi) = E(e^{i\phi X})$$

So,

$$e^{i\phi X(t)}$$

$$f(x, v, T, \phi) = E(\quad | x(t) = x, v(t) = v)$$

by varying the price in T.

By Fourier inversion :

$$P_j(x, V, t) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left[e^{-i\phi \ln(k)} \frac{f_j(x, v_1 t, \phi)}{i\phi} \right] d\phi \quad (18)$$

→ Then we are asking why,

$$f_j(x, v, T, \phi) = e^{i\phi x}$$

In fact, at maturity T, we are no longer looking for the expectation, we already have $x(T)$, the log of the underlying price at T, so:

$$E(e^{i\phi X(T)} | x(T)) = e^{i\phi x(T)}$$

→ At this stage, solving such an equation analytically is beyond the scope of our work:

$$f(x, v, T, \phi) = E(e^{i\phi X(t)} | x(t) = x, v(t) = v)$$

has an affine form according to martingale theory.

→ In other words we can write that:

$$f_j(x, v, t, \phi) = e^{C(T-t; \phi) + D(T-t; \phi)v + i\phi x}$$

→ We know that f_j checks (12)

We therefore just need to compute the various derivatives of f_j and substitute them into (12).

We set $h = T - t$.

$$\frac{\partial f}{\partial h} = f \left(\frac{\partial C}{\partial h} + v \frac{\partial D}{\partial h} \right)$$

$$\frac{\partial f}{\partial x} = i\phi f$$

$$\frac{\partial f}{\partial v} = Df$$

$$\frac{\partial^2 f}{\partial x^2} = -\phi^2$$

$$\frac{\partial^2 f}{\partial x \partial v} = i\phi \cdot Df$$

$$\frac{\partial^2 f}{\partial v^2} = D^2 f$$

So:

$$-\left(\frac{\partial C}{\partial h} + v \frac{\partial D}{\partial h}\right) + (h + u_j v)(i\phi) + (a - b_j v)D + \frac{1}{2}v(-\phi^2) + p\sigma v(i\phi \cdot D) + \frac{1}{2}\sigma^2 v D^2 = 0$$

Thus we find by separating the v terms from the others that:

$$C(h, \phi) = r\phi i h + \frac{a}{\sigma^2} \left[(bj - p\sigma\phi i + d)r - 2 \ln \left(\frac{1 - ge^{dh}}{1 - g} \right) \right]$$

$$D(h, \phi) = \frac{bj - p\sigma\phi i + d}{\sigma^2} \left(\frac{1 - e^{dh}}{1 - ge^{dh}} \right)$$

With

$$g = \frac{bj - P\sigma\phi i + d}{bj - p\sigma\phi i - d}$$

And

$$d = \sqrt{(p\sigma\phi i - bj)^2 - \sigma^2(2u_j\phi i - \phi^2)}$$

→ Then we know that:

$$C(s, v, t) = sP_1 - ke^{-r(T-t)} \cdot P_2$$

We replace P_1 and P_2 with the Fourier transforms calculated earlier, and then within these transforms, we substitute the f terms with the values computed in equation (17).

Python code B-S

```
import yfinance as yf

import numpy as np

def calcul_volatilite(ticker="TSLA", nb_jours=500):
    """
    Calcule la volatilité historique annualisée d'une action.
    """
    data = yf.download(ticker, period=f"{int(nb_jours*1.5)}d") # un peu plus large en cas de jours
    manquants
    data = data.dropna()
    data = data.tail(nb_jours)

    # Calcul des rendements journaliers log
    data['log_return'] = np.log(data['Close'] / data['Close'].shift(1))
    daily_vol = data['log_return'].std()

    annualized_vol = daily_vol * np.sqrt(252) # 252 jours de bourse/an
    return annualized_vol
```

```

import pandas as pd

import numpy as np

from scipy.stats import norm

# ----- PARAMÈTRES -----

spot_price = 318.38 # prix actuel de l'action Tesla

risk_free_rate = 0.05 # taux sans risque (ex: 5%)

time_to_maturity = 3/365 # en années (ex: 1 an)

implied_volatility = calcul_volatilite(ticker="TSLA", nb_jours=500) # Volatilité implicite réaliste pour Tesla

# ----- FORMULE DE BLACK-SCHOLES -----

def black_scholes_call(S, K, T, r, sigma):

    d1 = (np.log(S / K) + (r + 0.5 * sigma ** 2) * T) / (sigma * np.sqrt(T))

    d2 = d1 - sigma * np.sqrt(T)

    return S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)

# ----- CHARGEMENT ET TRAITEMENT -----

file_path = "C:/Users/julia/OneDrive/EDHEC/M1/Thesis/options_sp500.xlsx"

df_raw = pd.read_excel(file_path, header=None, skiprows=3)

# Sélectionner les colonnes contenant Strike, Bid, Ask

df_clean = df_raw[[7, 10, 11]].copy()

df_clean.columns = ['Strike', 'Bid', 'Ask']

```

```

df_clean = df_clean.dropna()

# Nettoyer les formats (au cas où)

for col in ['Strike', 'Bid', 'Ask']:

    df_clean[col] = df_clean[col].astype(str).str.replace(',', '').str.replace(' ', '').astype(float)

# ----- CALCUL DU PRIX BLACK-SCHOLES -----

df_clean['BS_Price'] = df_clean.apply(lambda row: black_scholes_call(

    S=spot_price,

    K=row['Strike'],

    T=time_to_maturity,

    r=risk_free_rate,

    sigma=implied_volatility

), axis=1)

# ----- AFFICHAGE -----

print(df_clean[['Strike', 'Bid', 'Ask', 'BS_Price']])

# Optionnel : sauvegarder le résultat

# df_clean.to_excel("BS_Tesla_Options.xlsx", index=False)

```

Python Code Heston model V1 for Tesla (Part 4)

```
import numpy as np
```

```

import pandas as pd

from scipy.integrate import quad

# Charger les données depuis Excel

file_path = "C:/Users/julia/OneDrive/EDHEC/M1/Thesis/options_sp500.xlsx"

df_raw = pd.read_excel(file_path, header=None, skiprows=3)

# Extraire Strike, Bid, Ask

df_clean = df_raw[[7, 10, 11]].copy()

df_clean.columns = ['Strike', 'Bid', 'Ask']

df_clean = df_clean.dropna()

# Nettoyage des données

for col in ['Strike', 'Bid', 'Ask']:

    df_clean[col] = df_clean[col].astype(str).str.replace(',', '').str.replace(' ', '').astype(float)

# Paramètres du modèle Heston (exemples réalistes pour Tesla)

S = 318.38      # prix spot

r = 0.05        # taux sans risque

T = 3           # maturité

v0 = calcul_volatilite(ticker="TSLA", nb_jours=500)

    # variance initiale (20%2)

kappa = 2       # vitesse de rappel

```

```

theta = 0.04 # variance à long terme

sigma = 0.4 # volatilité de la volatilité

rho = -0.7 # corrélation


# Fonction caractéristique de Heston

def heston_char_func(phi, Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

    x = np.log(S)

    a = kappa * theta

    u = 0.5 if Pnum == 1 else -0.5

    b = kappa - rho * sigma if Pnum == 1 else kappa

    d = np.sqrt((rho * sigma * phi * 1j - b)**2 - sigma**2 * (2 * u * phi * 1j - phi**2))

    g = (b - rho * sigma * phi * 1j + d) / (b - rho * sigma * phi * 1j - d)

    C = r * phi * 1j * T + (a / sigma**2) * ((b - rho * sigma * phi * 1j + d) * T - 2 * np.log((1 - g * np.exp(d
* T)) / (1 - g)))

    D = ((b - rho * sigma * phi * 1j + d) / sigma**2) * ((1 - np.exp(d * T)) / (1 - g * np.exp(d * T)))

    return np.exp(C + D * v0 + 1j * phi * x)


# Intégrale pour P1 et P2

def P_integral(Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

    integrand = lambda phi: np.real(np.exp(-1j * phi * np.log(K)) * heston_char_func(phi, Pnum, S, K, T, r,
kappa, theta, sigma, v0, rho) / (1j * phi))

    result, _ = quad(integrand, 1e-5, 100)

    return 0.5 + result / np.pi

```

```

# Prix Call via Heston

def heston_call_price(S, K, T, r, kappa, theta, sigma, v0, rho):

    P1 = P_integral(1, S, K, T, r, kappa, theta, sigma, v0, rho)

    P2 = P_integral(2, S, K, T, r, kappa, theta, sigma, v0, rho)

    return S * P1 - K * np.exp(-r * T) * P2


# Application aux strikes

df_clean['Heston_Call'] = df_clean['Strike'].apply(

    lambda K: heston_call_price(S, K, T, r, kappa, theta, sigma, v0, rho)

)


# Affichage des résultats

print(df_clean[['Strike', 'Bid', 'Ask', 'Heston_Call']].head())


# Optionnel : export

# df_clean.to_excel("heston_calls_tesla.xlsx", index=False)

```

Python Code best estimators (Part 4)

```

import numpy as np

import pandas as pd

from scipy.integrate import quad

from itertools import product

```

```

# ----- CHARGEMENT DU FICHIER -----

df = pd.read_excel("C:/Users/julia/OneDrive/EDHEC/M1/Thesis/option Metrics/option call.xlsx")

# ----- RENOMMER LES COLONNES -----

df = df.rename(columns={

    "strike": "Strike",

    "bid": "Bid",

    "ask": "Ask",

    "vol": "IV",

    "act_symbol": "Symbol",

    "Spot ": "Spot"

})

# ----- DATES ET TEMPS -----

df['date'] = pd.to_datetime(df['date'], errors='coerce')

df['expiration'] = pd.to_datetime(df['expiration'], errors='coerce')

df['T'] = (df['expiration'] - df['date']).dt.days / 365

# ----- NETTOYAGE -----

for col in ['Strike', 'Bid', 'Ask', 'IV', 'Spot']:

    df[col] = df[col].astype(str).str.replace(',', '').str.replace(' ', '').astype(float)

```

```

df['Market_Price'] = (df['Bid'] + df['Ask']) / 2

df['v0'] = df['IV'] ** 2

# ----- RÉDUCTION À 200 LIGNES -----

df = df.iloc[:200].copy()

# ----- MODÈLE DE HESTON -----

def heston_char_func(phi, Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

    x = np.log(S)

    a = kappa * theta

    u = 0.5 if Pnum == 1 else -0.5

    b = kappa - rho * sigma if Pnum == 1 else kappa

    d = np.sqrt((rho * sigma * phi * 1j - b)**2 - sigma**2 * (2 * u * phi * 1j - phi**2))

    g = (b - rho * sigma * phi * 1j + d) / (b - rho * sigma * phi * 1j - d)

    try:

        C = r * phi * 1j * T + (a / sigma**2) * ((b - rho * sigma * phi * 1j + d) * T - 2 * np.log((1 - g *
np.exp(d * T)) / (1 - g)))

        D = ((b - rho * sigma * phi * 1j + d) / sigma**2) * ((1 - np.exp(d * T)) / (1 - g * np.exp(d * T)))

    except:

        return np.nan

    return np.exp(C + D * v0 + 1j * phi * x)

def P_integral(Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

    try:

```



```

    integrand = lambda phi: np.real(

        np.exp(-1j * phi * np.log(K)) *

        heston_char_func(phi, Pnum, S, K, T, r, kappa, theta, sigma, v0, rho) / (1j * phi)

    )

    result, _ = quad(integrand, 1e-5, 100)

    return 0.5 + result / np.pi

except:

    return np.nan

def heston_call_price(S, K, T, r, kappa, theta, sigma, v0, rho):

    try:

        P1 = P_integral(1, S, K, T, r, kappa, theta, sigma, v0, rho)

        P2 = P_integral(2, S, K, T, r, kappa, theta, sigma, v0, rho)

        return S * P1 - K * np.exp(-r * T) * P2

    except:

        return np.nan

# ----- LISTES DE PARAMÈTRES -----

r = 0.02625

kappa_list = np.linspace(0, 10, 5)

theta_list = np.linspace(0, 1, 20)

sigma_list = np.linspace(0, 1, 5)

rho_list = np.linspace(-1, 1, 5)

```

```

param_grid = list(product(kappa_list, theta_list, sigma_list, rho_list))

total = len(param_grid)

# ----- CALCUL SÉQUENTIEL AVEC MESSAGES -----

results = []

for i, (kappa, theta, sigma, rho) in enumerate(param_grid):

    errors = []

    for _, row in df.iterrows():

        model_price = heston_call_price(

            S=row['Spot'], K=row['Strike'], T=row['T'], r=r,

            kappa=kappa, theta=theta, sigma=sigma, v0=row['v0'], rho=rho

        )

        market_price = row['Market_Price']

        if market_price > 0 and not np.isnan(model_price):

            error = abs((model_price - market_price) / market_price)

            errors.append(error)

    avg_error = np.mean(errors) if errors else np.inf

    print(f"✔ Combinaison {i+1}/{total} terminée : kappa={kappa:.2f}, theta={theta:.2f},
sigma={sigma:.2f}, rho={rho:.2f}, erreur={avg_error:.4f}")

    results.append((kappa, theta, sigma, rho, avg_error))

# ----- MEILLEURE COMBINAISON -----

best = min(results, key=lambda x: x[4])

```

```

print(f"\n✅ Meilleure combinaison : kappa={best[0]:.2f}, theta={best[1]:.2f}, sigma={best[2]:.2f},
rho={best[3]:.2f}")

print(f"📊 Erreur relative moyenne minimale : {best[4]*100:.2f} %")

```

Python Code B-S model V2 (Part 5)

```

import pandas as pd
import numpy as np
from scipy.stats import norm

# ----- FORMULE DE BLACK-SCHOLES -----
def black_scholes_call(S, K, T, r, sigma):
    if T <= 0 or sigma <= 0 or S <= 0:
        return np.nan
    d1 = (np.log(S / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return S * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)

# ----- CHARGEMENT DU FICHER -----
df = pd.read_excel("C:/Users/julia/OneDrive/EDHEC/M1/Thesis/option Metrics/option call.xlsx")

# ----- RENOMMER LES COLONNES -----
df = df.rename(columns={
    "strike": "Strike",
    "bid": "Bid",
    "ask": "Ask",
    "vol": "IV",
    "act_symbol": "Symbol",
    "Spot ": "Spot" # Corriger l'espace final dans le nom de colonne
})

# ----- CONVERSION DES DATES -----
df['date'] = pd.to_datetime(df['date'], errors='coerce')
df['expiration'] = pd.to_datetime(df['expiration'], errors='coerce')
df['T'] = (df['expiration'] - df['date']).dt.days / 365

# ----- NETTOYAGE -----

```

```

for col in ['Strike', 'Bid', 'Ask', 'IV', 'Spot']:
    df[col] = df[col].astype(str).str.replace('.', ',').str.replace(' ', '').astype(float)

# ----- PRIX MOYEN DU MARCHÉ -----
df['Market_Price'] = (df['Bid'] + df['Ask']) / 2

# ----- CALCUL DU PRIX BLACK-SCHOLES -----
risk_free_rate = 0.02625
df['BS_Price'] = df.apply(lambda row: black_scholes_call(
    S=row['Spot'],
    K=row['Strike'],
    T=row['T'],
    r=risk_free_rate,
    sigma=row['IV']
), axis=1)

# ----- COMPARAISON -----
df['Diff'] = df['BS_Price'] - df['Market_Price']
df['Relative_Error (%)'] = 100 * df['Diff'] / df['Market_Price']

# ----- AFFICHAGE FINAL -----
print(df[['Symbol', 'date', 'expiration', 'Strike', 'Spot', 'Market_Price', 'TV', 'T', 'BS_Price', 'Diff',
'Relative_Error (%)']])

df.to_excel("C:/Users/julia/OneDrive/EDHEC/M1/Thesis/B-S.xlsx", index=False)

```

Python Code B-S model V2 (Part 5)

```

import numpy as np

import pandas as pd

from scipy.integrate import quad

# ----- CHARGEMENT DU FICHER -----

df = pd.read_excel("C:/Users/julia/OneDrive/EDHEC/M1/Thesis/option Metrics/option call.xlsx")

```

```

# ----- RENOMMER LES COLONNES -----

df = df.rename(columns={

    "strike": "Strike",

    "bid": "Bid",

    "ask": "Ask",

    "vol": "IV",

    "act_symbol": "Symbol",

    "Spot ": "Spot" # Corrige l'espace après Spot

})

# ----- CONVERSION DES DATES -----

df['date'] = pd.to_datetime(df['date'], errors='coerce')

df['expiration'] = pd.to_datetime(df['expiration'], errors='coerce')

df['T'] = (df['expiration'] - df['date']).dt.days / 365

# ----- NETTOYAGE -----

for col in ['Strike', 'Bid', 'Ask', 'IV', 'Spot']:

    df[col] = df[col].astype(str).str.replace(',', '.').str.replace(' ', '').astype(float)

# ----- PRIX MOYEN MARCHÉ -----

df['Market_Price'] = (df['Bid'] + df['Ask']) / 2

```

```

# ----- PARAMÈTRES HESTON -----

r = 0.02625

kappa = 2    # vitesse de rappel

theta = 0.04 # variance à long terme

sigma = 0.4  # volatilité de la volatilité

rho = -0.7   # corrélation


# ----- VARIANCE INITIALE -----

df['v0'] = df['IV'] ** 2


# ----- FONCTIONS HESTON -----

def heston_char_func(phi, Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

    x = np.log(S)

    a = kappa * theta

    u = 0.5 if Pnum == 1 else -0.5

    b = kappa - rho * sigma if Pnum == 1 else kappa

    d = np.sqrt((rho * sigma * phi * 1j - b)**2 - sigma**2 * (2 * u * phi * 1j - phi**2))

    g = (b - rho * sigma * phi * 1j + d) / (b - rho * sigma * phi * 1j - d)

    C = r * phi * 1j * T + (a / sigma**2) * ((b - rho * sigma * phi * 1j + d) * T - 2 * np.log((1 - g * np.exp(d
* T)) / (1 - g)))

    D = ((b - rho * sigma * phi * 1j + d) / sigma**2) * ((1 - np.exp(d * T)) / (1 - g * np.exp(d * T)))

    return np.exp(C + D * v0 + 1j * phi * x)


def P_integral(Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

```

```

integrand = lambda phi: np.real(

    np.exp(-1j * phi * np.log(K)) *

    heston_char_func(phi, Pnum, S, K, T, r, kappa, theta, sigma, v0, rho) / (1j * phi)

)

result, _ = quad(integrand, 1e-5, 100)

return 0.5 + result / np.pi

def heston_call_price(S, K, T, r, kappa, theta, sigma, v0, rho):

    try:

        P1 = P_integral(1, S, K, T, r, kappa, theta, sigma, v0, rho)

        P2 = P_integral(2, S, K, T, r, kappa, theta, sigma, v0, rho)

        return S * P1 - K * np.exp(-r * T) * P2

    except:

        return np.nan

# ----- CALCUL DU PRIX HESTON -----

df['Heston_Call'] = df.apply(lambda row: heston_call_price(

    S=row['Spot'],

    K=row['Strike'],

    T=row['T'],

    r=r,

    kappa=kappa,

    theta=theta,

```

```

sigma=sigma,

v0=row['v0'],

rho=rho

), axis=1)

# ----- ERREUR RELATIVE -----

df['Relative_Error (%)'] = 100 * (df['Heston_Call'] - df['Market_Price']) / df['Market_Price']

# ----- AFFICHAGE FINAL -----

print(df[['Symbol', 'Strike', 'Spot', 'TV', 'Market_Price', 'Heston_Call', 'Relative_Error (%)']])

df.to_excel("C:/Users/julia/OneDrive/EDHEC/M1/Thesis/heston_calls.xlsx", index=False)

```

Python Code Heston V2 (Part 5)

```

#Code avec paramètre de base

import numpy as np

import pandas as pd

from scipy.integrate import quad

# ----- CHARGEMENT DU FICHIER -----

df = pd.read_excel("C:/Users/julia/OneDrive/EDHEC/M1/Thesis/option Metrics/option call.xlsx")

# ----- RENOMMER LES COLONNES -----

df = df.rename(columns={

```



```

"strike": "Strike",

"bid": "Bid",

"ask": "Ask",

"vol": "IV",

"act_symbol": "Symbol",

"Spot ": "Spot" # Corrige l'espace après Spot
}))

# ----- CONVERSION DES DATES -----

df['date'] = pd.to_datetime(df['date'], errors='coerce')

df['expiration'] = pd.to_datetime(df['expiration'], errors='coerce')

df['T'] = (df['expiration'] - df['date']).dt.days / 365

# ----- NETTOYAGE -----

for col in ['Strike', 'Bid', 'Ask', 'IV', 'Spot']:

    df[col] = df[col].astype(str).str.replace(',', '').str.replace(' ', '').astype(float)

# ----- PRIX MOYEN MARCHÉ -----

df['Market_Price'] = (df['Bid'] + df['Ask']) / 2

# ----- PARAMÈTRES HESTON -----

r = 0.02625

kappa = 10    # vitesse de rappel

```

```

theta = 0.0 # variance à long terme

sigma = 1 # volatilité de la volatilité

rho = -1 # corrélation


# ----- VARIANCE INITIALE -----

df['v0'] = df['IV'] ** 2


# ----- FONCTIONS HESTON -----

def heston_char_func(phi, Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

    x = np.log(S)

    a = kappa * theta

    u = 0.5 if Pnum == 1 else -0.5

    b = kappa - rho * sigma if Pnum == 1 else kappa

    d = np.sqrt((rho * sigma * phi * 1j - b)**2 - sigma**2 * (2 * u * phi * 1j - phi**2))

    g = (b - rho * sigma * phi * 1j + d) / (b - rho * sigma * phi * 1j - d)

    C = r * phi * 1j * T + (a / sigma**2) * ((b - rho * sigma * phi * 1j + d) * T - 2 * np.log((1 - g * np.exp(d
* T)) / (1 - g)))

    D = ((b - rho * sigma * phi * 1j + d) / sigma**2) * ((1 - np.exp(d * T)) / (1 - g * np.exp(d * T)))

    return np.exp(C + D * v0 + 1j * phi * x)


def P_integral(Pnum, S, K, T, r, kappa, theta, sigma, v0, rho):

    integrand = lambda phi: np.real(

        np.exp(-1j * phi * np.log(K)) *

        heston_char_func(phi, Pnum, S, K, T, r, kappa, theta, sigma, v0, rho) / (1j * phi)

```

```

)

result, _ = quad(integrand, 1e-5, 100)

return 0.5 + result / np.pi


def heston_call_price(S, K, T, r, kappa, theta, sigma, v0, rho):

    try:

        P1 = P_integral(1, S, K, T, r, kappa, theta, sigma, v0, rho)

        P2 = P_integral(2, S, K, T, r, kappa, theta, sigma, v0, rho)

        return S * P1 - K * np.exp(-r * T) * P2

    except:

        return np.nan


#supprimer les 2/3 premières valeurs :

df = df.iloc[int(len(df) * 2 / 3):].copy()


# ----- CALCUL DU PRIX HESTON -----

df['Heston_Call'] = df.apply(lambda row: heston_call_price(

    S=row['Spot'],

    K=row['Strike'],

    T=row['T'],

    r=r,

    kappa=kappa,

    theta=theta,

```

```

sigma=sigma,

v0=row['v0'],

rho=rho

), axis=1)

# ----- ERREUR RELATIVE -----

df['Relative_Error (%)'] = 100 * np.abs((df['Heston_Call'] - df['Market_Price']) / df['Market_Price'])

# ----- AFFICHAGE FINAL -----

print(df[['Symbol', 'Strike', 'Spot', 'TV', 'Market_Price', 'Heston_Call', 'Relative_Error (%)']])

# ----- MOYENNE DE L'ERREUR RELATIVE -----

mean_relative_error = df['Relative_Error (%)'].dropna().mean()

print(f"\n📊 Moyenne de l'erreur relative : {mean_relative_error:.2f} %")

print(f"📉 Médiane de l'erreur :", df['Relative_Error (%)'].median())

#Sauvegarder Excel

df.to_excel("C:/Users/julia/OneDrive/EDHEC/M1/Thesis/heston_calls.xlsx", index=False)

```